

Molecular Biomarkers in Localized Prostate Cancer: ASCO Guideline Summary

Scott E. Eggener, MD¹; R. Bryan Rumble, MSc²; and Himisha Beltran, MD³

The clinical practice guideline on molecular biomarkers in localized prostate cancer¹ evaluated evidence to provide physicians, including medical oncologists, radiation oncologists, and urologists; other health care practitioners (eg, nurses, nurse practitioners, physician assistants); patients; and caregivers with recommendations regarding the role of molecular diagnostics in localized prostate cancer.

Prostate cancer is the most commonly diagnosed cancer in men in the United States (approximately 174,650 in 2019), nearly 20% of all new cancers, and

the second leading cause of cancer-related death (approximately 31,620 in 2019).² At diagnosis, there is a diverse spectrum of clinical courses that range from indolent features with a negligible likelihood of morbidity or mortality to characteristics reflecting near certitude of eventual metastases and cancer-specific death. Predicting future clinical behavior is imperfect but constitutes the foundation of physician counseling and patient management decisions. Potential ramifications of various management strategies on quality of life and cancer-specific outcomes are highly variable and often

TABLE 1. Definitions and Terms used in the Guideline

Molecular biomarker	A biological <i>molecule</i> found in blood, body fluid, or tissue that provides information regarding the presence or absence of a disease, prognosis, or likelihood to respond to a specific treatment. Molecular biomarkers broadly encompass DNA, transcriptome, protein, metabolic, and other tissue-based or cellular biomarkers.
Cellular biomarker	Cells found in blood, body fluid, or tissue that provide information regarding the presence or absence of a disease, prognosis, or likelihood to respond to a specific treatment
Genomic biomarker	Genetic material (including DNA or RNA) found in blood, body fluid, or tissue that provides information regarding the presence or absence of a disease, prognosis, or likelihood to respond to a specific treatment
Active surveillance	Monitoring prostate cancer rather than immediately treating it
Clinically significant prostate cancer	Various definitions; in general, clinical, pathologic, or biomarker features of prostate cancer suggesting a possibility of becoming clinically relevant (symptoms or metastases)
Low-risk prostate cancer	Gleason score ≤ 6 (grade group 1), prostate-specific antigen (PSA) < 10 ng/mL, and nonpalpable or only palpable in less than half of one lobe of the prostate (clinical stage T1c or T2a)
Intermediate-risk prostate cancer	PSA 10-20 ng/mL, a clinical stage of T2b-T2c, or Gleason score 7 (grade group 2 to 3) without meeting any criteria for high risk
High-risk prostate cancer	Clinical stage T3a, Gleason score ≥ 8 (grade group ≥ 4), or PSA ≥ 20 ng/mL
Very high-risk prostate cancer	Clinical stage T3b-T4, any primary Gleason pattern 5 (grade group 5) or $>$ four cores with Gleason score 8 to 10 (grade group ≥ 4)
Assay validity	Comprehensive experiments that evaluate and document the quantitative performance of an assay, including sensitivity, specificity, accuracy, precision, reproducibility, detection limit, range, and limits of quantitation.
Analytic validity	How well a test predicts the presence or absence of a particular disease, condition, or state
Clinical validity	How well the test result is related to the presence, absence, or risk of a specific disease.
Clinical utility	The ability of a screening or diagnostic test to prevent or ameliorate adverse health outcomes such as mortality, morbidity, or disability
Tumor heterogeneity	Variability among cancer cells, including cellular morphology, genomic alterations, gene expression, metabolism, proliferation, and metastatic potential
Predictive biomarker	Provides information regarding response to a specific treatment
Diagnostic biomarker	Provides information regarding the likelihood of disease presence or absence
Prognostic biomarker	Provides information about the patient's overall cancer outcome (eg, disease recurrence, progression, death) independent of treatment received

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on December 10, 2019 and published at ascopubs.org/journal/op on January 28, 2020; DOI <https://doi.org/10.1200/JOP.19.00752>

Reprint Requests: 2318 Mill Road, Suite 800, Alexandria, VA 22314; guidelines@asco.org

With panel representation from the American Urological Association (AUA), the College of American Pathologists (CAP), and European Association of Urology (EAU). At the time of publication, this ASCO Guideline has been endorsed by the EAU.

THE BOTTOM LINE**Molecular Biomarkers in Localized Prostate Cancer: ASCO Guideline****Guideline Question**

Are there molecular biomarkers with utility in the management of localized prostate cancer?

Target Population

Men with localized prostate cancer.

Target Audience

Medical oncologists, radiation oncologists, urologists, other health care practitioners, patients, and caregivers.

Methods

An Expert Panel was convened to develop clinical practice guideline recommendations based on a systematic review of the medical literature.

Recommendations

Summary. Numerous molecular biomarkers have been developed to improve risk stratification and patient management. Few panels have undergone extensive validation; however, five are commercially available and have been shown in retrospective analyses to provide additional information beyond standard clinical models in prognostication or patient selection for therapy. While these tissue-based tests may improve risk stratification when added to standard clinical parameters, we recommend considering their use in situations in which the assay result, when considered as a whole with routine clinical factors, is likely to have an impact on patient management. Examples include select men with high-volume low-risk or favorable intermediate-risk prostate cancer who are considering active surveillance or in men with high-risk features for treatment intensification (see Table 3 in the original guideline:¹ Summary Box Recommendations 1-4). While testing may influence management decisions, there is no high-level evidence that the results from these panels will improve quality of life or cancer-specific outcomes. Additional biomarkers have been evaluated that do not have sufficient data to be clinically actionable or are not commercially available. We recommend continued investigation of tissue-based molecular biomarkers in the context of clinical trials.

Clinical Question 1

Are there molecular biomarkers to identify patients with prostate cancer who are most likely to benefit from active surveillance?

Recommendation 1.1. Commercially available molecular biomarkers (ie, Oncotype Dx Prostate, Prolaris, Decipher, and ProMark) may be offered in situations in which the assay result, when considered as a whole with routine clinical factors, is likely to have an impact on patient management. Routine ordering of molecular biomarkers is not recommended (Type: evidence-based; Evidence quality: intermediate; Strength of recommendation: moderate).

Recommendation 1.2. Any additional molecular biomarkers evaluated do not have sufficient data to be clinically actionable or are not commercially available, and thus should not be offered (Type: evidence-based; Evidence quality: insufficient; Strength of recommendation: moderate).

Clinical Question 2

Are there molecular biomarkers to diagnose clinically significant prostate cancer?

Recommendation 2.1. Commercially available molecular biomarkers (ie, Oncotype Dx Prostate, Prolaris, Decipher, and ProMark) may be offered in situations in which the assay result, when considered as a whole with routine clinical factors, is likely to have an impact on patient management. Routine ordering of molecular biomarkers is not recommended (Type: evidence-based; Evidence quality: intermediate; Recommendation: moderate).

Recommendation 2.2. Any additional molecular biomarkers evaluated do not have sufficient data to be clinically actionable or are not commercially available, thus should not be offered (Type: evidence-based; Evidence quality: insufficient; Strength of recommendation: moderate).

Clinical Question 3

Are there molecular biomarkers to guide the decision between postprostatectomy adjuvant versus salvage radiation?

(continued on following page)

THE BOTTOM LINE (CONTINUED)

Recommendation 3.1. The Expert Panel recommends consideration of a commercially available molecular biomarker (eg, Decipher Genomic Classifier) in situations in which the assay result, when considered as a whole with routine clinical factors, is likely to have an impact on patient management. In the absence of prospective clinical trial data, routine use of genomic biomarkers in the postprostatectomy setting to determine adjuvant versus salvage radiation or to initiate systemic therapies should not be offered (Type: evidence-based; Evidence quality: intermediate; Strength of recommendation: moderate).

Recommendation 3.2. Any additional molecular biomarkers evaluated do not have sufficient data to be clinically actionable or are not commercially available, and thus should not be offered (Type: evidence-based; Evidence quality: insufficient; Strength of recommendation: moderate).

Clinical Question 4

What are the comparative strengths and weaknesses of genomics versus MRI in identifying clinically significant prostate cancer?

Recommendation 4. In men with newly diagnosed prostate cancer who are eligible for active surveillance, both genomics and MRI intend to identify clinically significant cancers. The Expert Panel endorses their use only in situations in which the result, when considered as a whole with routine clinical factors, is likely to have an impact on patient management. This may include, for instance, the initial management of men potentially eligible for active surveillance, in whom each of these approaches may provide clinically relevant and actionable information. These tests may provide information independent of routine clinical parameters and independent of one another (Type: informal consensus; benefits:harms ratio unknown; Evidence quality: low; Strength of recommendation: weak).

Additional Resources

More information, including a supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/genitourinary-cancer-guidelines. Patient information is available at www.cancer.net

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

profound. Risk stratification has traditionally relied upon serum prostate-specific antigen (PSA), Gleason grading, and clinical stage to classify localized cancers as low, intermediate, or high risk. By using additional biopsy data (eg, total number or percentage of positive cores), more nuanced risk stratification can be achieved (eg, Cancer of the Prostate Risk Assessment [CAPRA] score, nomograms).^{3,4}

A variety of molecular biomarkers have been developed, evaluated, and commercialized with an overarching aim to further personalize risk stratification, more comprehensively inform management decisions, and consequently improve the quality of care. Biomarkers may be assessed from blood, body fluid, or tumor tissue with the intention of adding clinically actionable data independent of information previously attained (ie, PSA, grade group, clinical stage). The goal of a biomarker is to achieve any, or ideally all, of the following: estimate the likelihood of a disease characteristic being present or absent, determine prognosis more accurately, or provide the probability of response to a specific treatment.

To evaluate localized prostate cancer biomarker studies to date, ASCO formed an Expert Panel that included representatives from the American Urological Association (AUA), the College of American Pathologists (CAP), and the

European Association of Urology (EAU). The potential role of molecular biomarkers in four clinical scenarios was addressed: determining patients who are most likely to benefit from surveillance of localized prostate cancer, identifying prostate cancers with the potential to ultimately cause symptoms or develop metastases, deciding between adjuvant or salvage radiation therapy after radical prostatectomy, and assessing the relative value of genomics versus magnetic resonance imaging (MRI). The purpose of the clinical practice guideline was to provide recommendations that are based on the best available evidence regarding the role of molecular, cellular, and genomic biomarkers in localized prostate cancer. Because these biomarkers are relatively new and the field is rapidly evolving, future directions are also discussed. [Table 1](#) provides operational definitions for the terms used within this guideline.

Additional information is available at www.asco.org/genitourinary-cancer-guidelines. Patient information is available at www.cancer.net.

WHAT IS PRACTICE CHANGING

The new ASCO guideline provides guidance regarding the role of tissue-based molecular biomarkers in localized

prostate cancer for personalizing risk stratification and informing management decisions. This includes identifying appropriate patients for active surveillance, selecting patients for adjuvant versus salvage radiation, or predicting adverse pathologic, biochemical, or cancer-specific survival

outcomes. Molecular data could potentially improve confidence in management choice, especially in gray zone areas where clinical variables are less certain, although high-quality prospective evidence, particularly in a randomized design, is currently lacking.

AFFILIATIONS

¹University of Chicago Medicine, Chicago, IL

²American Society of Clinical Oncology, Alexandria, VA

³Dana-Farber Cancer Institute, Boston, MA

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Road, Suite 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST AND DATA AVAILABILITY STATEMENT

Disclosures provided by the authors and data availability statement (if applicable) are available with this article at DOI <https://doi.org/10.1200/JOP.19.00752>.

AUTHOR CONTRIBUTIONS

Conception and design: All authors

Administrative support: All authors

Collection and assembly of data: All authors

Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

Molecular Biomarkers in Localized Prostate Cancer: ASCO Guideline was developed and written by Scott E. Eggener, R. Bryan Rumble, Andrew J. Armstrong, Todd M. Morgan, Tony Crispino, Philip Cornford, Theodorus van der Kwast, David J. Grignon, Alex J. Rai, Neeraj Agarwal, Eric A. Klein, Robert B. Den, and Himisha Beltran.

REFERENCES

1. Eggener SE, Rumble RB, Armstrong AJ, et al: Molecular biomarkers in localized prostate cancer: ASCO guideline. *J Clin Oncol* 38:1474-1494, 2020
2. Siegel RL, Miller KD, Jemal A: Cancer statistics, 2019. *CA Cancer J Clin* 69:7-34, 2019
3. Cooperberg MR, Pasta DJ, Elkin EP, et al: The University of California, San Francisco Cancer of the Prostate Risk Assessment score: A straightforward and reliable preoperative predictor of disease recurrence after radical prostatectomy. *J Urol* 173:1938-1942, 2005
4. Stephenson AJ, Scardino PT, Eastham JA, et al: Preoperative nomogram predicting the 10-year probability of prostate cancer recurrence after radical prostatectomy. *J Natl Cancer Inst* 98:715-717, 2006

Gain Recognition as an ASCO Journal Reviewer

Participate as a journal reviewer and play an integral role in maintaining the quality and integrity of ASCO journals. Benefits and recognition for reviewing ASCO journals content include:

- Staying up to date on the latest oncology research
- Increased exposure to key leaders in oncology
- Career advancement opportunities
- ASCO members who become reviewers can earn FASCO points

Get started today at ascopubs.org/reviewers

ASCO Journals



AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Molecular Biomarkers in Localized Prostate Cancer: ASCO Guideline Summary

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/op/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](https://openpayments.gov)).

Scott E. Eggerer

Consulting or Advisory Role: Sophiris Bio, Francis Medical, InSightec, Profound Medical

Speakers' Bureau: Janssen

Travel, Accommodations, Expenses: Janssen Biotech, InSightec, Sophiris Bio

Uncompensated Relationships: Steba Biotech

R. Bryan Rumble

Employment: Park Lane Terrace (I)

Himisha Beltran

Consulting or Advisory Role: Janssen Oncology, Genzyme, GlaxoSmithKline, AbbVie, Astellas Pharma, AstraZeneca

Research Funding: Janssen Pharmaceuticals (Inst), AbbVie (Inst) Stemcentrx (Inst), Eli Lilly (Inst)

Travel, Accommodations, Expenses: Janssen Oncology

No other potential conflicts of interest were reported.